

Rounak Sonavane

AI Systems Engineer | Intelligent Automation | Robotics | Computer Vision

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Profile

AI Systems Engineer who designs and ships end-to-end intelligent systems across LLM agents, machine learning, computer vision and robotics — from model training through API design to automated deployment and live monitoring.

Tech Stack

Languages	Python, C++ , C, SQL
AI & LLM Systems	LLM Agents, LangGraph, LangChain, RAG, Fine-Tuning, Transformers, PyTorch, TensorFlow
Robotics & Perception	ROS2, MoveIt, ros2_control, URDF, ESP32, Computer Vision, YOLO, OpenCV, Whisper
Backend & Data	FastAPI, Flask, REST APIs, React, PostgreSQL, MongoDB, Vector Search (ChromaDB, FAISS)
MLOps & Infrastructure	Docker, CI/CD (GitHub Actions), Git, AWS, Linux, Cloudflare, Tailscale, n8n, Observability

Projects

ClauseGuard — AI Legal Contract Review Platform clauseguard.rounak.co | github.com/Rounak7721/ClauseGuard | Aug 2026

Python, FastAPI, PyTorch, Transformers, LangChain, LangGraph, RAG, Groq, ChromaDB, React, TypeScript, Docker, CI/CD

- Built and deployed a production contract-review platform classifying clauses into 41 CUAD categories with a fine-tuned Legal-BERT model, paired with a deterministic rule-based risk engine that keeps risk scoring reproducible and auditable
- Raised classifier macro-F1 from 0.63 to 0.78 and accuracy from 0.79 to 0.86 by root-causing a training-label conflict and switching checkpoint selection from accuracy to F1
- Engineered a LangGraph tool-calling agent producing RAG-grounded, citation-backed clause reports, with interrupt/resume human-in-the-loop confirmation gating expensive operations
- Automated delivery via GitHub Actions CI/CD building multi-architecture images to GHCR and deploying over Tailscale to a Cloudflare-tunnelled host, with independent n8n uptime monitoring on AWS

MindDesk — Multimodal Agentic AI Platform

github.com/Rounak7721/MindDesk | Mar 2026 – Jun 2026

Python, Flask, React, LangChain, RAG, Ollama, YOLO, OpenCV, Whisper, PostgreSQL, MongoDB, FAISS, ChromaDB, REST APIs

- Architected a custom plan-and-execute orchestration engine coordinating 17 specialized tools through a decoupled registry, with typed plan validation, retry-repair loops and cross-step result chaining
- Built a document intelligence pipeline for semantic retrieval across 4 document formats using FAISS and ChromaDB
- Unified computer vision, speech processing, document intelligence and IoT automation into one locally-hosted platform with per-task model routing and role-based access control

IRIS — Intelligent Robotic Perception and Manipulation System

github.com/Rounak7721/iris | Jan 2026 – Present

Python, C++ , ROS2, MoveIt, ros2_control, URDF, PyTorch, YOLO, OpenCV, Whisper, Qt, ESP32, Linux

- Developing a custom 6-DOF robotic manipulator across 11 ROS2 packages — URDF modelling, a C++ ros2_control hardware interface driving servos over serial, MoveIt motion planning, and ESP32 firmware
- Built a dual-camera perception stack separating overview tracking from eye-in-hand YOLO detection, plus a local voice pipeline using wake-word detection, VAD and Whisper transcription

Agentic Lead Qualification Automation

github.com/Rounak7721/ai-lead-qualification-system | Apr 2026

n8n, LLMs, LLaMA, Groq, Ollama, JavaScript, REST APIs, Webhooks, Airtable, Slack, Google Sheets

- Built an automated lead capture, LLM scoring, tiering and routing pipeline with idempotent notification flags preventing duplicate alerts across real-time and scheduled workflows
- Separated live CRM state from an append-only audit log, with hybrid cloud and local LLM deployment for privacy

Virtual Mouse — Gesture-Controlled HCI System

github.com/Rounak7721/Virtual-Mouse | Oct 2024

Python, OpenCV, MediaPipe, NumPy, Real-Time Computer Vision

- Built a touchless HCI system mapping 21-point MediaPipe hand landmarks to cursor, click, scroll and drag actions

Experience

Self-Directed Engineering — Robotics & Applied AI Projects

Jun 2025 – Present

- Self-taught the ROS2 stack — MoveIt, ros2_control, SLAM — and applied it to a ground-up 6-DOF arm build
- Moved into agentic AI and production delivery — a hand-built agent engine, then a live CI/CD-deployed platform

AI Intern — Ramkrishna Foundry Pvt. Ltd.

Jan 2025 – Apr 2025

- Developed a company website using Vite, JavaScript and Express with an integrated LLM-powered chatbot
- Automated contact form and data capture workflows via Google Sheets integration and web analytics, and prototyped LLM-powered features for internal business workflows

Education & Additional

B.Tech in Artificial Intelligence & Machine Learning — Sanjay Ghodawat University

Jun 2025

Open To: AI Systems Engineer, Intelligent Automation, Robotics Software, Applied AI, Computer Vision

Languages: English, Hindi, Marathi